English Premier League Stats (2021-2025)

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2026-03-24

Introduction: This project explores match performance in the English Premier League from 2021 to 2024. The goal is to analyze home field advantage, team performance metrics, and match outcomes using descriptive statistics and visualizations.

```
##Load Data
```

```
##Fix Column Names
```

```
#Make clean new column names
colnames(data) <- data[1, ]

#Remove the first row
data <- data[-1, ]

#Reset row index
rownames(data) <- NULL

#Check
summarize(.) = dplyr.summarize(data)

# Check updated structure
str(data)
```

```
## 'data.frame':    1140 obs. of  18 variables:
##  $ date      : chr "29th May 2022" "29th May 2022" "29th May 2022" "29th May 2022" ...
##  $ time      : chr "16:00" "16:00" "16:00" "16:00" ...
##  $ stadium   : chr "Emirates Stadium" "Villa Park" "Bank Community Stadium" "Stamford Bridge" ...
##  $ class     : chr "H" "H" "H" "H" ...
##  $ attendance: chr "60,260" "42,618" "11,120" "40,120" ...
##  $ home_team : chr "Arsenal" "Aston Villa" "Brentford" "Chelsea" ...
##  $ h_score   : chr "3" "2" "1" "1" ...
##  $ a_score   : chr "1" "1" "2" "0" ...
##  $ away_team : chr "Watford" "Chelsea" "Crystal Palace" "Leeds" ...
##  $ h_possess : chr "62" "41" "47" "52" ...
##  $ a_possess : chr "38" "59" "53" "48" ...
##  $ h_shots   : chr "21" "11" "12" "14" ...
##  $ a_shots   : chr "5" "14" "10" "11" ...
##  $ h_goals   : chr "3" "2" "1" "1" ...
##  $ a_goals   : chr "1" "1" "2" "0" ...
##  $ h_corners : chr "7" "3" "4" "6" ...
##  $ a_corners : chr "2" "7" "3" "4" ...
##  $ h_fouls   : chr "9" "12" "11" "9" ...
##  $ a_fouls   : chr "12" "9" "9" "11" ...
##  $ h_offside : chr "3" "2" "2" "1" ...
##  $ a_offside : chr "1" "3" "1" "2" ...
##  $ h_yellows : chr "1" "2" "2" "1" ...
##  $ a_yellows : chr "2" "1" "1" "2" ...
##  $ h_reds    : chr "0" "0" "0" "0" ...
##  $ a_reds    : chr "0" "0" "0" "0" ...
```

```
#Convert Data Types
```

```
# Convert key columns to numeric
datafootball$h_shots <- as.numeric(datafootball$h_shots)
datafootball$a_shots <- as.numeric(datafootball$a_shots)
datafootball$h_goals <- as.numeric(datafootball$h_goals)
datafootball$a_goals <- as.numeric(datafootball$a_goals)
datafootball$h_possession <- as.numeric(datafootball$h_possession)
datafootball$a_possession <- as.numeric(datafootball$a_possession)

# Fix attendance column (remove commas)
datafootball$attendance <- as.numeric(gsub(",", "", datafootball$attendance))
```

```
##Create Match Outcome Variables
```

```
##
## Away Win  Draw Home Win
##  380  380  380
```

```
##Home Vs Away Win Percentage
```

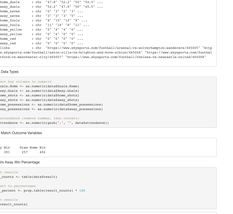
```
# Create counts
result_counts <- table(datafootball)

# Convert to percentage
result_percent <- prop.table(result_counts) * 100

# Print results
print(result_counts)
```

```
##
## Away Win  Draw Home Win
##  380  380  380
```

```
print(result_percent)
```

```
##
## Away Win  Draw Home Win
## 33.3333 33.3333 33.3333
```

```
ggplot(data=datafootball,
       main = "Match Outcomes",
       xlab = "Outcome", "Count", "Grey")
```



The bar chart shows the distribution of match outcomes across the dataset. Home wins appear to occur more frequently than away wins and draws, suggesting a potential home field advantage in the Premier League. The relatively high number of draws also indicates that many matches end in ties, which may reflect that away teams remain competitive, but they do not occur as often as home wins. Draws are the least frequent outcome, showing that most matches tend to produce a clear winner rather than ending evenly. Overall, this pattern supports the idea that playing at home may provide a slight advantage, but matches are still relatively competitive across all outcomes.

```
##Goal Analysis
```

```
mean(datafootball$a_goals, na.rm = TRUE)
```

```
## [1] 1.365263
```

```
mean(datafootball$h_goals, na.rm = TRUE)
```

```
## [1] 1.294201
```

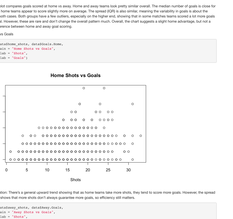
```
ggplot(data=datafootball, data=data_shots,
       aes(x = a) + geom_bar(),
       main = "Goals Scored: Home vs Away",
       ylab = "Goals")
```

Interpretation:

This boxplot compares goals scored at home vs away. Home and away teams look pretty similar overall. The median number of goals is close for both, but home teams appear to score slightly more on average. The boxes (IQR) are also similar, meaning the variability in goals is about the same for both home and away teams. The whiskers show the typical range of goals, and home teams tend to reach a little more goals than usual. Overall, there aren't big differences between home and away goals, but home teams do have a slight edge in scoring. Overall, this chart suggests a slight home advantage, but not a huge difference between home and away goal scoring.

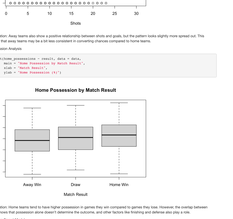
```
##Shots vs Goals
```

```
ggplot(data=data_shots, data=data_shots,
       aes(x = h_shots, y = h_goals),
       xlab = "Shots",
       ylab = "Goals")
```



Interpretation: There's a general upward trend showing that as home teams take more shots, they tend to score more goals. However, the spread of points shows that more shots don't always guarantee more goals, so efficiency still matters.

```
ggplot(data=data_shots, data=data_shots,
       aes(x = a_shots, y = a_goals),
       xlab = "Shots",
       ylab = "Goals")
```



Interpretation: Away teams also show a positive relationship between shots and goals, but the pattern looks slightly more spread out. This suggests that away teams may be a bit less consistent in converting chances compared to home teams.

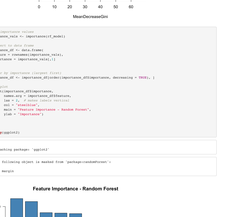
```
##Possession Analysis
```

```
boxplot(home_possession ~ result, data = data,
        main = "Home Possession by Match Result",
        xlab = "Match Result",
        ylab = "Home Possession (%)")
```

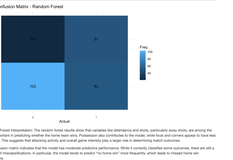
Interpretation: Home teams tend to have higher possession in games they win compared to games they lose. However, the overlap between groups shows that possession alone doesn't determine the outcome, and other factors like finishing and defense also play a role.

```
##Random Forest Model
```

```
# Load packages
library(randomForest)

## randomForest 4.7-1.1

## Type 1(Trees) to see new features/changes/bug fixes.

# Convert binary outcome (home win = 1, other = 0)
datafootball_win_binary <- ifelse(datafootball == "Home Win", 1, 0)
datafootball_win_binary <- as.factor(datafootball_win_binary)

# Select features
features <- data[, c("home_possession",
                     "away_possession",
                     "home_shots",
                     "away_shots",
                     "home_goals",
                     "away_goals",
                     "home_corners",
                     "away_corners",
                     "attendance")]

# Convert to numeric (keep column names)
features$home_possession <- as.numeric(features$home_possession)
features$away_possession <- as.numeric(features$away_possession)
features$home_shots <- as.numeric(features$home_shots)
features$away_shots <- as.numeric(features$away_shots)
features$home_goals <- as.numeric(features$home_goals)
features$away_goals <- as.numeric(features$away_goals)
features$home_corners <- as.numeric(features$home_corners)
features$away_corners <- as.numeric(features$away_corners)
features$attendance <- as.numeric(gsub(",", "", features$attendance))

# Combine
model_data <- cbind(features, home_win_binary = datafootball_win_binary)

# Split into 2 sets
set.seed(123)
n <- nrow(model_data)
train_indices <- sample(1:n, .5 * n)
train_data <- model_data[train_indices, ]
test_data <- model_data[-train_indices, ]

# Train model
set.seed(123)
rf_model <- randomForest(
  home_win_binary ~ .,
  data = train_data,
  ntree = 500,
  importance = TRUE
)

# Predictions
y_pred <- predict(rf_model, X_test)

# Accuracy
accuracy <- sum(y_pred == y_test) / length(y_test)
print(paste("Accuracy:", round(accuracy, 3)))

## [1] "Accuracy: 0.676"

# Confusion matrix
cm <- table(Predicted = y_pred, Actual = y_test)
print("Confusion Matrix:")

## [1] "Confusion Matrix:"

print(cm)

##       Actual
## Predicted   0   1
##         0 103  51
##         1  51 103

# Feature importance
importance <- importance(rf_model)
print("Feature Importance:")

## [1] "Feature Importance:"

print(importance_sorted)

##               MeanDecreaseGini
## home_possession         11.6423
## away_possession         12.0731
## home_shots              14.4205
## away_shots              14.4023
## home_fouls              10.5734
## away_fouls              10.4204
## home_corners            10.4154
## away_corners            10.5116
## attendance              17.2187

# Plot feature importance
varImpPlot(rf_model)
```




```
# Get importance values
importance_scores <- importance(rf_model)

# Convert to data frame
importance_DF <- data.frame(
  Feature = rownames(importance_scores),
  Importance = importance_scores[,1]
)

# Order by importance (largest first)
importance_DF <- importance_DF[order(importance_DF$Importance, decreasing = TRUE), ]

# Bar plot
ggplot(data=importance_DF$Importance,
       aes(x=reorder(Feature, -Importance),
           y = Importance),
       main = "Feature Importance - Random Forest",
       xlab = "Feature",
       ylab = "Importance")

library(ggplot2)

##
## Attaching package: 'ggplot2'

## The following object is masked from 'package:randomForest':
##
##     margin

# Plot
ggplot(data=df, aes(x = Actual, y = Predicted, fill = Freq)) +
  geom_tile() +
  geom_text(aligned(aes(label = Freq)) +
  ggplot(ggtitle("Confusion Matrix - Random Forest") +
  theme_minimal()
```




```
ds_df <- as.data.frame(ds)

ggplot(ds_df, aes(x = Actual, y = Predicted, fill = Freq)) +
  geom_tile() +
  geom_text(aligned(aes(label = Freq)) +
  ggplot(ggtitle("Confusion Matrix - Random Forest") +
  theme_minimal()
```


Random Forest Interpretation: The random forest results show that variables like attendance and shots, particularly away shots, are among the most important in predicting whether the home team wins. Possession also contributes to the model, while fouls and corners appear to have less influence. This suggests that shots and overall game intensity play a larger role in determining match outcomes.

The confusion matrix indicates that the model has moderate predictive performance. While it correctly classifies some outcomes, there are still a noticeable number of misclassifications. In particular, the model tends to predict "no home win" more frequently, which leads to missed home win predictions.

Overall, the model captures some meaningful patterns in the data, but its accuracy is limited. This suggests that adding more relevant features or further refining the model could help improve its performance.